Problem Formulation

Define two classes: class I and Class II

let $d(X)$ be a discriminant function = $W^T X$

if $X \in$ class I then $d(X) > 0$

if $X \in$ Class II then $d(X) < 0$

For misclassified X :

If $X \in$ class I and misclassified as Class II then $d(X)$ will be <0

If $X \in$ Class II and misclassified as Class I then $d(X)$ will be >0

Define Error function as:

$$\text{Error} = \sum_{\text{all misclassified vectors}(i)} W^T X_i \quad \Rightarrow \quad (W^T X_i) \text{ take positive and negative values}$$

X

Define δ_i as :

If $X \in$ class I and misclassified as Class II then $d(X)$ will be $<0 \Rightarrow \delta_i = -1 \Rightarrow d(X)\,\delta_i > 0$

If $X \in$ Class II and misclassified as Class I then $d(X)$ will be $>0 \Rightarrow \delta_i = +1 \Rightarrow d(X)\,\delta_i > 0$